

CRC Package Interface Specification

Current API design

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1 Overview

The package is centred around a single high-level function:

```
crc_fit(  
  data,  
  captures,  
  capture_formula,  
  outcome = NULL,  
  outcome_formula = NULL,  
  outcome_dist = c("ztnegbin", "ztpois"),
```

```

    misclass = NULL,
    latent_classes = 1L,
    control = NULL
)

```

The current implementation focuses on **validation and parsing**. Model fitting (EM algorithm) will be implemented later.

2 Main arguments

Argument	Description
<code>data</code>	One row per observed unit.
<code>captures</code>	One-sided formula defining capture indicators.
<code>capture_formula</code>	Log-linear capture model. <code>.latent</code> denotes the latent class.
<code>outcome</code>	NULL, a single outcome variable, or <code>c(lower=..., upper=...)</code> .
<code>outcome_formula</code>	Mean model for the outcome distribution.
<code>outcome_dist</code>	Currently "ztnegbin" or "ztpois".
<code>misclass</code>	Misclassification specification for categorical variables.
<code>latent_classes</code>	Number of latent classes.
<code>control</code>	Numerical control options (reserved for future use).

3 Current workflow

```

crc_fit()

  match.arg(outcome_dist)
  validate_crc_input()
  parse_captures()
  parse_outcome()
  parse_misclass()
  parse_capture_formula()
  parse_outcome_formula()

```

At this stage the function returns an **unfitted model object** containing parsed specifications.

4 Parser responsibilities

4.1 `parse_captures()`

- validates the capture formula;
- verifies binary capture indicators;
- checks missing values;
- checks that every unit is observed by at least one source;
- returns a `crc_captures` object.

4.2 `parse_outcome()`

Supports:

- `NULL`;
- one exact outcome column;
- interval/right-censored outcomes via `lower/upper`.

Returns a standardized table with:

- `lower`
- `upper`
- `status`

4.3 `parse_capture_formula()`

Validates:

- one-sided formula;
- intercept;
- untransformed variables only;
- presence of all capture main effects;
- hierarchical interactions;
- `.latent`;
- allowed interactions:
 - `capture × capture`,
 - `capture × .latent`.

Returns a `crc_capture_formula` object.

4.4 `parse_outcome_formula()`

Validates:

- one-sided formula;
- intercept;
- categorical predictors;
- hierarchical interactions;
- optional use of `.latent`.

Returns a `crc_outcome_formula` object.

4.5 `parse_misclass()`

Expected structure:

```
misclass = list(  
  occupation = list(  
    matrix = Pi_occ,  
    true_if = ~ source_1  
  ),  
  contract = list(  
    matrix = Pi_contract  
  )  
)
```

Responsibilities:

- validates confusion matrices;
- validates category levels;
- validates `true_if`;
- constructs logical vectors identifying observations with known true category;
- returns a `crc_misclass` object.

5 Returned object

Currently `crc_fit()` returns an object of classes

```
c("crcfit_unfitted", "crcfit")
```

containing:

- parsed capture specification,
- parsed outcome specification,
- parsed capture model,
- parsed outcome model,
- parsed misclassification specification,
- model metadata.

6 Next implementation steps

1. Build internal model matrices.
2. Initialize parameters.
3. Implement the EM algorithm.
4. Compute standard errors.
5. Add `summary()`, `print()`, and `predict()` methods.

CRC Package Interface Specification

Model-matrix, initialization, E-step, and outcome-likelihood implementation addendum

1 Model-matrix infrastructure

The package now converts parsed model specifications into complete capture-cell grids and aligned observation-level states. The infrastructure includes the capture and outcome design matrices, misclassification contributions, and a direct mapping from each observation state to its capture cell. Construction of these components is integrated into `crc_fit()`, although model fitting has not yet been implemented.

1.1 Capture-model matrix

`build_capture_matrix()` constructs the Cartesian product of all capture histories, categorical levels used by the capture model, and latent classes. For misclassified variables, true-category levels and their ordering are obtained from the corresponding confusion matrices. The function returns:

- `cells`, the complete capture-model cell grid;
- `matrix`, the design matrix generated from the parsed terms;
- `unobserved`, an indicator of all-zero capture histories.

With J capture sources, K latent classes, and categorical variables with $C_1, \dots, C_p$ levels, the grid contains $2^J K \prod_{q=1}^p C_q$ rows.

1.2 Initial observation states

`build_observation_states()` repeats every observed unit once for each latent class. It retains the variables needed by the capture model, outcome model, and misclassification mechanisms. It also keeps aligned copies of the standardized outcome and observed values of misclassified variables. An observation identifier maps every expanded row back to its original unit. Only variables used by the capture or outcome model are retained in the state data. Recorded values of variables subject to misclassification are stored separately for evaluation of their confusion-matrix contributions.

1.3 Misclassified-variable expansion

`expand_misclass_states()` expands the current states over every possible true level of one misclassified variable. Each expanded row receives the contribution

$$\Pr(\tilde{G}_j = h \mid G_j = g)$$

from its confusion matrix. When `true_if` applies, the observed category receives probability one and all other candidate categories receive probability zero.

For several misclassified variables, separate confusion matrices are used and their contributions are multiplied. This corresponds to conditional independence of the misclassification mechanisms given their respective true categories. Dependent mechanisms may be supported in a future extension. These contributions are likelihood terms, not posterior probabilities, and must not be normalized at this stage.

`expand_all_misclass_states()` applies this expansion successively to every specified variable. When no misclassification mechanism is supplied, it assigns a contribution of one to every latent state.

1.4 Outcome-model matrix

`build_outcome_matrix()` applies the parsed outcome terms to the fully expanded state data. Its rows therefore remain aligned with the state data, standardized outcomes, observation identifiers, and accumulated misclassification contributions. It returns `NULL` when no outcome model is specified.

Every predictor in `outcome_formula` is required to occur in `capture_formula`. This ensures that the predictor is represented in the complete capture-cell grid and is consequently available for prediction in the all-zero cells. The formulas may contain different terms; only the set of variables is constrained.

1.5 Capture-cell indices

`build_capture_index()` returns one integer index for every expanded observation state. Matching uses the capture indicators, categorical variables in the capture model, and latent class. If the i th value is r , row i of the state data uses row r of the capture-cell grid and capture design matrix. The vector avoids repeated joins during future EM iterations. Construction fails if any state cannot be matched to the grid.

1.6 Unified model-matrix object

`build_model_matrices()` coordinates the complete construction sequence and returns an object of class `crc_model_matrices` with:

- `capture`, containing the cell grid, capture design matrix, and all-zero-cell indicator;
- `states`, containing expanded state data and aligned metadata;
- `outcome_matrix`, containing the outcome design matrix or `NULL`;
- `capture_index`, linking state rows to capture-cell rows.

`crc_fit()` constructs this object after parsing and compatibility validation and stores it as `model_matrices`. The returned object remains in class `crcfit_unfitted` and has `fitted = FALSE`.

1.7 Cross-component validation

Compatibility checks now reject outcome predictors absent from the capture model. They also reject variables specified in `misclass` that are used by neither the capture nor the outcome model. This prevents redundant state expansion and makes the population-cell interpretation explicit.

2 Initialization

`initialize_crc()` creates normalized initial weights for the expanded observation states and expected counts for the complete capture-cell grid. For multiple latent classes, independent gamma draws produce a symmetric Dirichlet allocation within each observation and candidate true-category configuration. These allocations are multiplied by the misclassification contributions and normalized within each observed unit. The concentration is set by `control$init_alpha` and defaults to 20.

State weights are aggregated through the precomputed capture-cell indices. Observed cells receive the corresponding weighted counts, while every all-zero capture cell receives an initial expected count of one. The resulting `crc_initialization` object stores the state weights, cell counts, and Dirichlet concentration and is retained by `crc_fit()`.

3 Expectation step

`expectation_crc()` implements the distribution-independent part of the E-step. It accepts one fitted capture-model mean for every row of the complete capture-cell grid and, when an outcome model is present, one outcome log-likelihood contribution for every expanded observation state. A missing outcome model is represented by zero log-likelihood contributions, equivalent to likelihood contributions of one.

For state (i, g, k) , the unnormalized log weight is

$$\log m_{Y_{i,g,k}} + \log \Pr(\tilde{G}_i \mid G_i = g) + \log L_V(i \mid g).$$

Exact zero misclassification probabilities remain negative infinity on the log scale and therefore produce posterior weights of exactly zero. The function verifies that each observed unit has at least one admissible state and normalizes the weights within unit after subtracting the largest log weight. This log-sum-exp calculation avoids unnecessary underflow and overflow.

The normalized state weights are aggregated through `capture_index` to update the expected counts for observed capture histories. Expected counts for all-zero histories are set to their current fitted capture-model means. The function returns a `crc_expectation` object containing one posterior weight per expanded state and one expected count per capture cell. It is not yet called by `crc_fit()`, because the capture and outcome M-steps needed to supply its inputs have not been implemented.

3.1 Log-scale probability subtraction

`log_sub_crc()` provides the stable operation

$$\log\{\exp(a) - \exp(b)\}, \quad a \geq b,$$

using `expm1()`. This helper is intended for interval-censored outcome likelihoods, where the required probability is obtained by subtracting two survival probabilities. It also handles the case in which both probabilities are zero and their log values are negative infinity.

4 Outcome log-likelihood

`outcome_loglik_crc()` evaluates one outcome log-likelihood contribution for every expanded observation state. Its arguments are the aligned standardized outcome table, a positive mean `mu` for every state, the outcome distribution, and, for the negative binomial model, a single positive dispersion parameter `sigma`. Predictors in `outcome_formula`, including `.latent`, may therefore produce different means for different candidate states of the same observed unit.

Let W denote the corresponding untruncated count. Under "ztpois", $W \sim \text{Poisson}(\mu)$. Under "ztneqbin", W has mean μ , size $1/\sigma$, and variance $\mu + \sigma\mu^2$. In both cases the modeled outcome is the conditional variable $V = W \mid W \geq 1$. If f , F , and $S(q) = \Pr(W > q)$ denote the untruncated mass, distribution, and survival functions, respectively, the implemented contribution is

$$\ell_V = \begin{cases} \log f(v) - \log S(0), & \text{exact observation,} \\ \log\{S(l-1) - S(u)\} - \log S(0), & \text{interval } l \leq V \leq u, \\ \log S(l-1) - \log S(0), & \text{right censoring } V \geq l, \\ 0, & \text{missing outcome.} \end{cases}$$

Thus, interval bounds are inclusive and missing outcomes contribute a likelihood factor of one under the ignorable coarsening assumption. Exact contributions use the log probability mass directly. Interval contributions use upper-tail log probabilities and `log_sub_crc()`, avoiding unstable subtraction of nearly equal ordinary probabilities in the right tail. No artificial probability floor is applied.

5 Automated tests

The model-matrix, initialization, E-step, and outcome-likelihood infrastructure is covered by automated tests implemented with the `tinytest` package. Separate test files cover model matrices, misclassification expansion, cross-component compatibility, initialization, expectation calculations, and outcome log-likelihood calculations.

The existing tests cover capture grids and design matrices, row alignment, factor levels, all-zero cells, misclassification expansion, compatibility, initialization, and E-step behavior. Matrix, compatibility, and initialization tests exercise the implemented behavior through `crc_fit()`; the E-step is tested directly because it is not yet integrated into the fitting loop.

Initialization tests verify normalized, reproducible state weights and their capture-cell aggregation, including the single-class case and invalid control values. Fifteen E-step assertions cover posterior normalization, impossible states, cell aggregation, numerical stability, dimensions, and validation.

The outcome-likelihood tests compare the Poisson and negative binomial results with direct probability calculations for mixed exact, interval-censored, right-censored, and missing outcomes. They also verify inclusive interval bounds, a right-censoring boundary at one, numerical stability in the far right tail, and validation of the mean and dispersion inputs. The complete suite now contains 66 passing assertions, of which nine exercise `outcome_loglik_crc()`.

6 Required next steps

The model-matrix, initialization, distribution-independent E-step, and outcome-likelihood infrastructure and their automated tests are complete at the design level. Development should continue in the following order.

1. Implement separate M-step updates for the capture and outcome models, using the posterior state weights and the existing alignment indices.
2. Integrate initialization, both M-steps, outcome likelihood evaluation, and `expectation_crc()` into the EM iteration.
3. Add convergence controls, numerical safeguards, and diagnostics before exposing fitted estimates through user-facing methods.

EM updates must follow the methodology in the accompanying paper and simulation implementation. The existing automated tests establish the initialization, matrix, and alignment invariants on which those iterative components depend.